



INSTITUTE OF INFORMATION TECHNOLOGY
JAHANGIRNAGAR UNIVERSITY

3rd Year 1st Semester B.Sc. (Hons.) Final Examination 2021

COURSE CODE: IT-3101

COURSE TITLE: Operating Systems

TOTAL MARKS: 60

TIMES: 3 Hours

ANSWER ANY FIVE (5) QUESTIONS

1. a) "Operating system works as a resource allocator as well as program controller" - Justify the statement. 2.0
- b) How does the direct memory access (DMA) technique work? Explain with a proper diagram. 3.0
- c) Briefly describe the Dual-mode operation of an operating system. 22-1-c 3.0
- d) Define system call. Suppose you want to copy the contents of an existing file to a new file. Explain the process with a system call. 22-1-d 4.0
2. a) How address binding of instructions and data to memory addresses can happen? 4.0
- b) OS is responsible for the protection of the user's memory. This is accomplished by hardware. Draw the hardware and describe its operation principle. 4.0
- c) Describe the general dynamic storage allocation problem which concerns how to satisfy a request of size n from a list of free holes. 4.0
3. a) Write the types of execution that concurrent processes can perform. 3.0
- b) What are the benefits of using cooperating process? How Interprocess communication is performed? 22-2-b 4.0
- c) Write the solution goals of critical section problems in process synchronization. Mr. Mahmud has proposed a solution for process synchronization to access the critical section. Justify the solution goals and problems of the solution (if any): 5.0

bool LOCK = false; // shared

```
while (true)
{
    do non_critical_section
    while (LOCK == true) {}
    LOCK = true
    do critical_section
```

LOCK = false

4. a) Consider the following set of processes with the CPU burst time, arrival time, and priority. 9.0

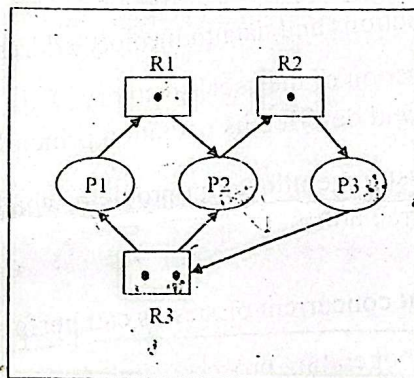
Process	Arrival Time	Burst Time	Priority
P1	5	8	4
P2	0	5	1
P3	2	3	2
P4	0	3	5
P5	2	2	3

Draw three Gantt charts illustrating the execution of the processes using the following algorithms. Also, calculate the average waiting time and average turnaround time for each of the algorithms.

- Preemptive Priority Scheduling
- Preemptive Shortest Job First
- Round Robin with Time Quantum=3

- b) Define the starvation problem and the convoy effect. How these can be solved? 3.0

5. a) How do you determine deadlock in a resource allocation graph? Does the following resource allocation graph contain a deadlock? Justify your answer. 4.0



- b) Considering the following snapshot of a system, answer the questions using the banker's algorithm: 8.0

Process	Allocation				Max				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P0	0	0	1	2	0	0	1	2	1	5	2	0
P1	1	0	0	0	1	7	5	0				
P2	1	3	5	4	2	3	5	6				
P3	0	6	3	2	0	6	5	2				
P4	0	0	1	4	0	6	5	6				

- Find the total number of instances in the system for each of the resources.
- What is the content of the need matrix?
- Is the system in a safe state? If yes, find the safe sequence.
- If a request from process P1 arrives for (0, 4, 2, 3), can the request be granted immediately? Justify your answer.

6. a) Consider the following page reference string: 1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3. Compare the performance for the following page replacement algorithms assuming three frames. 4.0

- Least Recently Used Algorithm
- Optimal Algorithm

- b) How would first-fit and best-fit algorithms place the following set of processes in order? 6.0

Process	Size	Turnaround	Block	Size
P1	212 k	3	B1	100 k
P2	417 k	1	B2	500 k
P3	112 k	2	B3	200 k
P4	426 k	1	B4	300 k

- c) Consider the following segment table: 2.0

Segment	Base	Limit
0	219	600
1	2300	14
2	90	100
3	1327	580
4	1957	96

What are the physical addresses for the following logical addresses?

- i) 0, 430 ii) 1, 10 iii) 2, 250 iv) 4, 112

7. a) Define fragmentation. Compare internal fragmentation with external fragmentation. 3.0

- b) Suppose that a disk drive has 2500 cylinders numbered 0 to 2499. The drive is currently serving a request at cylinder 2150. The queue of pending requests in order is: 6.0

20, 69, 4, 212, 2, 296, 800, 544, 618, 356, 523, 965, 36, 81

Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the requests for the following disk scheduling algorithms:

(i) SCAN

(ii) C-LOOK

(iii) SSTF (7586)

- c) What is the purpose of disk scheduling? You have to select an algorithm for disk scheduling. Identify the criteria you will follow for selecting a proper disk scheduling algorithm. 3.0



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INSTITUTE OF INFORMATION TECHNOLOGY

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3rd Year 1st Semester B.Sc. (Hons.) Final Examination 2021

COURSE CODE: IT-3103 (COMPUTER NETWORK AND INTERNET TECHNOLOGY)

TOTAL MARKS: 60

TIMES: 3 Hours

ANSWER ANY FIVE (5) QUESTIONS

1. a) Consider an application that transmit data at a steady rate (for example, the sender generates an N-bit unit of data every k time units, where k is small and fixed). Also, when such an application starts, it will continue running for a relatively long period of time. Answer the following questions, briefly justifying your answer: 6
 - i) Would a packet-switched network or a circuit-switched network be more appropriate for the application? Why?
 - ii) Suppose that a packet-switched network is used and the only traffic in the network comes from such applications as described above. Furthermore, assume that the sum of the application data rate is less than the capacities of each and every link. Is some form of congestion control needed? Why?
- b) Performance at a node is often measured not only in terms of delay, but also in terms of the probability of packet loss.....How.....Justify. 4
- c) Suppose users share a 2 Mbps link. Also suppose each user transmits continuously at 1 Mbps when transmitting, but each user transmit only 20 percent of the time. 2
 - i) When circuit switching is used, how many users can be supported?
 - ii) Suppose packet switching is used. Why will there be essentially no queuing delay before the link if two or fewer users transmit at the same time?
2. a) What is the advantage of DSL over Dial-up Internet access? 4
- b) Consider a highway that has a tollbooth every 100 km. Suppose that cars travel on the highway at a rate of 100 km/hour and maintain that speed between tollbooths. Suppose next that 10 cars, travelling together as a caravan, follow each other in a fixed order. Also suppose that each tollbooth services a car at a rate of one car per 12 seconds. Finally suppose whenever the first car of the caravan arrives at a tollbooth, it waits at the entrance until the other nine car of the caravan arrives at the tollbooth. 4
 - I) What time will each tollbooth require to push the entire caravan onto the highway?
 - II) In this analogy which one is transmission delay and which one is propagation delay?
- c) According to the definition, Facebook Messenger should follow P2P architecture but it cannot follow fully this architecture. Why? Explain. 4

P2P

3. a) Briefly describe what problem arises when http works with non-persistent connection. 4
- b) Distinguish between "data element" and "signal element" with an analogy. What is the relationship between data rate and signal rate? A signal is carrying data in which four data element is encoded as three signal element ($r = 4/3$). If the bit rate is 1 Mbps, what is the average value of baud rate if c is between 0 and 1? 4
- c) Why Web-based e-mail became popular when IMAP and POP3 were already there in the market? Who first implemented the concept of Web-based e-mail? POP3 4
4. a) TCP servers use different socket for each connecting client. Therefore, if there are n simultaneous connections, each from a different client host, how many socket would the TCP server need? 3
- b) Suppose that a Web server runs in Host C on port 80. Suppose this Web Server uses persistent connections and is currently receiving requests from two different hosts A and B. Are all of the requests being sent through the same socket at host C? If they are being passed through different sockets, do both of the sockets have port 80? Discuss and explain. 5
- c) In rdt protocols why do we need to introduce timer and sequence number? Explain. 4
5. a) Why error may be introduced during data transmission? What do you mean by redundancy and why it is added to a message? 2
- b) List some error-detection methods. Suppose, a transmitter wants to transmit the following text to a receiver: 4

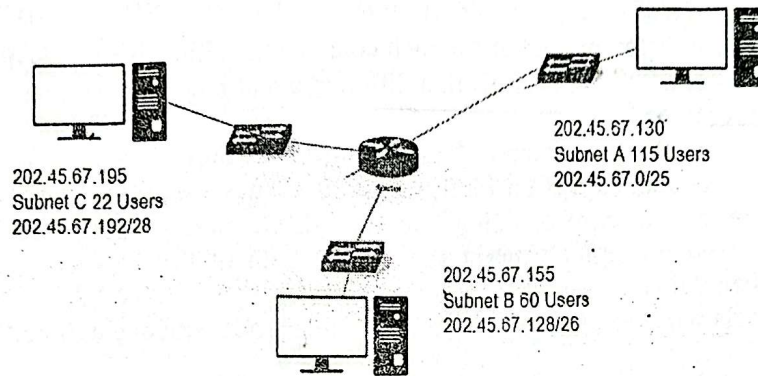
KB KHAN

Illustrate the internet checksum method to identify whether the message is accepted or rejected by the receiver. ASCII values for the given characters are:

Characters	ASCII Value	
	(decimal)	(Hex. decimal)
A	65	41
S	83	53
I	73	49
F	70	46
E	70	46
Space	32	20
K	75	4B
H	72	48
A	65	41
N	78	4E

- c) A code scheme has a Hamming distance $d_{\min} = 4$. What is the error detection and correction capability of this scheme? What will be the minimum Hamming distance for detecting 2-errors? 2
- d) A transmitter is transmitting an 8-bit codeword to a receiver using 4-bit even parity. If the received codeword is 001110010100, then determine whether error has occurred. 4

6. a) What is the difference between the role of a routing protocol from the role of a router itself? 2
- b) Why administrative distance is important for path selection? Explain with example. 4
- c) You are designing a network with subnet of variable sizes as shown below. Figure out two problems of your design for which you will not be able to communicate from one subnet to another. 4



- d) Suppose you are designing a network and after configuring each 2 path/subnet/router/switch you are finally trying to troubleshoot. List any four commands that can be used for this troubleshooting.
7. a) What is private ip address? Identify which one are public and private ip address of the following: 4
- 11.28.240.40 *pb*
 - 190.34.3.8 *pb*
 - 10.57.8.6 *pr*
 - 192.168.12.12 *pr*
 - 210.16.12.32 *pb*
 - 117.34.3.8 *pb*
- b) What restrictions are followed for classless address block? Suppose a block of 4 addresses is granted to a small organization. One of the addresses of the block is 205.16.37.39/28. Determine the 4
- First address in the block
 - Last address in the block
 - Total number of addresses in the block
- c) What do you mean by netid and hosted? How many valid networks and how many hosts in each network may exist in class A type? 2
- d) Define subnetting and supernetting. 2

0 - 15
16 - 31
32 - 47
48

127
191
223
231



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3rd Year 1st Semester B.Sc. (Hons.) Final Examination 2021

COURSE CODE: IT-3109

COURSE TITLE: PRINCIPLES OF ECONOMICS

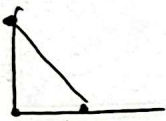
TOTAL MARKS: 60

TIME: 3 Hours

ANSWER ANY FIVE (5) QUESTIONS

1. a) Suppose there are six workers in an economy. A worker can produce either five cakes or three shirts per day. 12.0

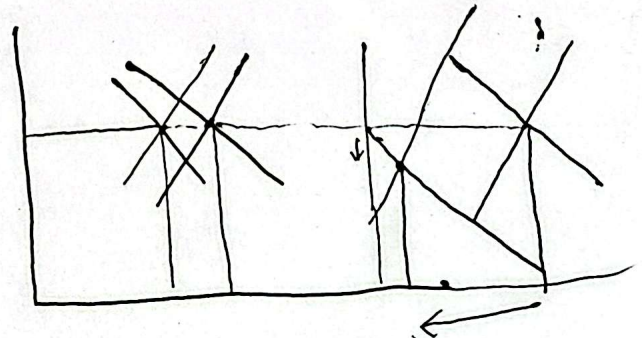
- Draw society's PPF.
- Why points inside the PPF are inefficient?
- Why points outside the PPF are unattainable? Explain.
- Why PPF shifts outward to the right?
- Why PPF is straight line in this case?



2. a) 'Scarcity implies choice; choice implies opportunity cost'-explain. 4.0
- b) What are the resources in the economy? Who are the owners of these resources? What are the rewards of these resources in production? 4.0
- c) Explain the characteristics of 'land' and 'labor' as factors of production. 4.0

3. a) Using supply and demand graphs and assuming competitive markets, show and explain the effect on equilibrium price and quantity of the following: 12.0
- Increased prices of milk and sugar on the market for an ice-cream cone.
 - Incidence of bird flu on the market for chicken.
 - A hot weather on the market for ice cream cones.
 - Bumper production on the market for rice.

4. a) State the law of Diminishing Marginal Product. ✓ 4.0
- b) Define marginal cost (MC) and average total cost (ATC). Explain the relationship between MC 4.0
- c) Why ATC curve is U-shaped? 4.0



5.

Suppose in a hypothetical economy, three goods are produced: Mango, Banana and Chicken. The quantity of output and prices of mango, banana and chicken are given in the following Table.

	Output			Price, Tk/Unit		
	2014	2015	2016	2014	2015	2016
Mango	150	160	165	105	110	115
Banana	510	515	530	7	8	10
Chicken	315	320	325	110	115	125

- a) Define nominal and real GDP. Calculate the nominal GDP for the year 2014, 2015 and 2016. Suppose, 2015 is the base year, and then calculate the real GDP for the year 2014, 2015 and 2016. Define growth rate. Calculate the growth rate of real GDP for the year 2015 and 2016. 6.0

- b) Define GDP deflator. What does it measure? According to the GDP deflator, how much prices have increased for the year 2015 and 2016? 6.0

6. a) Define price elasticity of demand. Briefly explain its determinants? 5.0

- b) On Tuesday, price and quantity demanded are Tk. 7 and 120 units, respectively. Ten days later, price and quantity demanded are Tk. 5 and 150 units, respectively. Calculate the price elasticity of demand and interpret the result. 7.0

7. a) Why is there structural unemployment? Explain. 6.0

- b) Suppose the population of Bangladesh is 180 million. Of which 80% are adult. The population not in labor force is 40 million and the number of employed is 90 million. Calculate the labor force participation rate and unemployment rate. 6.0